arXiv:1807.03748v2 [cs.LG] 22 Jan 2019

Representation Learning with Contrastive Predictive Coding

Aaron van den Oord
DeepMind
avdnoord@google.com

Yazhe Li
DeepMind
yazhe@google.com

Oriol Vinyals
DeepMind
vinyals@google.com

Abstract

While supervised learning has enabled great progress in many applications, unsupervised learning has not seen such widespread adoption, and remains an important and challenging endeavor for artificial intelligence. In this work, we propose a universal unsupervised learning approach to extract useful representations from high-dimensional data, which we call Contrastive Predictive Coding. The key insight of our model is to learn such representations by predicting the future in *latent* space by using powerful autoregressive models. We use a probabilistic contrastive loss which induces the latent space to capture information that is maximally useful to predict future samples. It also makes the model tractable by using negative sampling. While most prior work has focused on evaluating representations for a particular modality, we demonstrate that our approach is able to learn useful representations achieving strong performance on four distinct domains: speech, images, text and reinforcement learning in 3D environments.

1 Introduction

Learning high-level representations from labeled data with layered differentiable models in an end-to-end fashion is one of the biggest successes in artificial intelligence so far. These techniques made manually specified features largely redundant and have greatly improved state-of-the-art in several real-world applications [1, 2, 3]. However, many challenges remain, such as data efficiency, robustness or generalization.

Improving representation learning requires features that are less specialized towards solving a single supervised task. For example, when pre-training a model to do image classification, the induced features transfer reasonably well to other image classification domains, but also lack certain information such as color or the ability to count that are irrelevant for classification but relevant for e.g. image captioning [4]. Similarly, features that are useful to transcribe human speech may be less suited for speaker identification, or music genre prediction. Thus, unsupervised learning is an important stepping stone towards robust and generic representation learning.

Despite its importance, unsupervised learning is yet to see a breakthrough similar to supervised learning: modeling high-level representations from raw observations remains elusive. Further, it is not always clear what the ideal representation is and if it is possible that one can learn such a representation without additional supervision or specialization to a particular data modality.

One of the most common strategies for unsupervised learning has been to predict future, missing or contextual information. This idea of predictive coding [5, 6] is one of the oldest techniques in signal processing for data compression. In neuroscience, predictive coding theories suggest that the brain predicts observations at various levels of abstraction [7, 8]. Recent work in unsupervised learning has successfully used these ideas to learn word representations by predicting neighboring words [9]. For images, predicting color from grey-scale or the relative position of image patches has also been

Preprint. Work in progress.

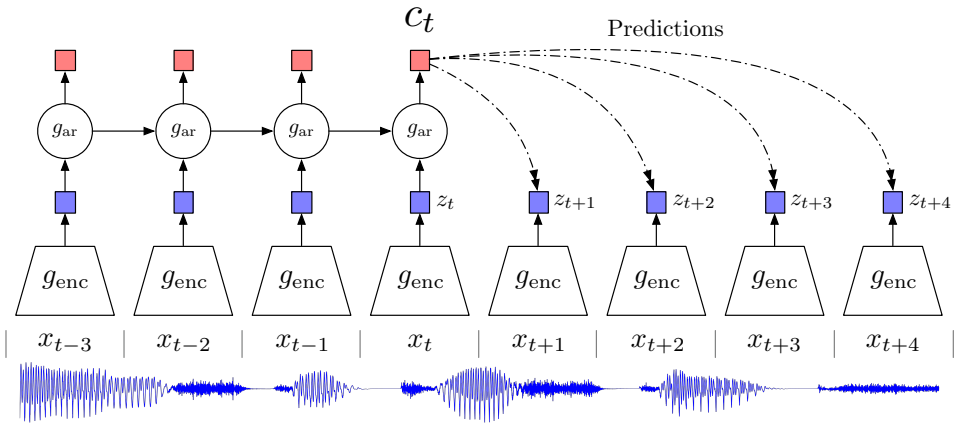


Figure 1: Overview of Contrastive Predictive Coding, the proposed representation learning approach. Although this figure shows audio as input, we use the same setup for images, text and reinforcement learning.

shown useful [10, 11]. We hypothesize that these approaches are fruitful partly because the context from which we predict related values are often conditionally dependent on the same shared high-level latent information. And by casting this as a prediction problem, we automatically infer these features of interest to representation learning.

In this paper we propose the following: first, we compress high-dimensional data into a much more compact latent embedding space in which conditional predictions are easier to model. Secondly, we use powerful autoregressive models in this latent space to make predictions many steps in the future. Finally, we rely on Noise-Contrastive Estimation [12] for the loss function in similar ways that have been used for learning word embeddings in natural language models, allowing for the whole model to be trained end-to-end. We apply the resulting model, Contrastive Predictive Coding (CPC) to widely different data modalities, images, speech, natural language and reinforcement learning, and show that the same mechanism learns interesting high-level information on each of these domains, outperforming other approaches.

2 Contrastive Predicting Coding

We start this section by motivating and giving intuitions behind our approach. Next, we introduce the architecture of Contrastive Predictive Coding (CPC). After that we explain the loss function that is based on Noise-Contrastive Estimation. Lastly, we discuss related work to CPC.

2.1 Motivation and Intuitions

The main intuition behind our model is to learn the representations that encode the underlying shared information between different parts of the (high-dimensional) signal. At the same time it discards low-level information and noise that is more local. In time series and high-dimensional modeling, approaches that use next step prediction exploit the local smoothness of the signal. When predicting further in the future, the amount of shared information becomes much lower, and the model needs to infer more global structure. These 'slow features' [13] that span many time steps are often more interesting (e.g., phonemes and intonation in speech, objects in images, or the story line in books.).

One of the challenges of predicting high-dimensional data is that unimodal losses such as mean-squared error and cross-entropy are not very useful, and powerful conditional generative models which need to reconstruct every detail in the data are usually required. But these models are computationally intense, and waste capacity at modeling the complex relationships in the data x, often ignoring the context c. For example, images may contain thousands of bits of information while the high-level latent variables such as the class label contain much less information (10 bits for 1,024 categories). This suggests that modeling $p(x|c)$ directly may not be optimal for the purpose of extracting shared information between x and c. When predicting future information we instead encode the target x (future) and context c (present) into a compact distributed vector representations (via non-linear

learned mappings) in a way that maximally preserves the mutual information of the original signals x and c defined as

$$I(x;c) = \sum_{x,c} p(x,c) \log \frac{p(x|c)}{p(x)}. \tag{1}$$

By maximizing the mutual information between the encoded representations (which is bounded by the MI between the input signals), we extract the underlying latent variables the inputs have in commmon.

2.2 Contrastive Predictive Coding

Figure 1 shows the architecture of Contrastive Predictive Coding models. First, a non-linear encoder g_{enc} maps the input sequence of observations x_t to a sequence of latent representations $z_t = g_{\text{enc}}(x_t)$, potentially with a lower temporal resolution. Next, an autoregressive model g_{ar} summarizes all $z_{\leq t}$ in the latent space and produces a context latent representation $c_t = g_{\text{ar}}(z_{\leq t})$.

As argued in the previous section we do not predict future observations x_{t+k} directly with a generative model $p_k(x_{t+k}|c_t)$. Instead we model a density ratio which preserves the mutual information between x_{t+k} and c_t (Equation 1) as follows (see next sub-section for further details):

$$f_k(x_{t+k}, c_t) \propto \frac{p(x_{t+k}|c_t)}{p(x_{t+k})} \tag{2}$$

where $\propto$ stands for 'proportional to' (i.e. up to a multiplicative constant). Note that the density ratio f can be unnormalized (does not have to integrate to 1). Although any positive real score can be used here, we use a simple log-bilinear model:

$$f_k(x_{t+k}, c_t) = \exp\left(z_{t+k}^T W_k c_t\right), \tag{3}$$

In our experiments a linear transformation $W_k^T c_t$ is used for the prediction with a different W_k for every step k. Alternatively, non-linear networks or recurrent neural networks could be used.

By using a density ratio $f(x_{t+k}, c_t)$ and inferring z_{t+k} with an encoder, we relieve the model from modeling the high dimensional distribution x_{t_k}. Although we cannot evaluate $p(x)$ or $p(x|c)$ directly, we can use samples from these distributions, allowing us to use techniques such as Noise-Contrastive Estimation [12, 14, 15] and Importance Sampling [16] that are based on comparing the target value with randomly sampled negative values.

In the proposed model, either of z_t and c_t could be used as representation for downstream tasks. The autoregressive model output c_t can be used if extra context from the past is useful. One such example is speech recognition, where the receptive field of z_t might not contain enough information to capture phonetic content. In other cases, where no additional context is required, z_t might instead be better. If the downstream task requires one representation for the whole sequence, as in e.g. image classification, one can pool the representations from either z_t or c_t over all locations.

Finally, note that any type of encoder and autoregressive model can be used in the proposed framework. For simplicity we opted for standard architectures such as strided convolutional layers with resnet blocks for the encoder, and GRUs [17] for the autoregresssive model. More recent advancements in autoregressive modeling such as masked convolutional architectures [18, 19] or self-attention networks [20] could help improve results further.

2.3 InfoNCE Loss and Mutual Information Estimation

Both the encoder and autoregressive model are trained to jointly optimize a loss based on NCE, which we will call InfoNCE. Given a set $X = \{x_1, \dots x_N\}$ of N random samples containing one positive sample from $p(x_{t+k}|c_t)$ and $N-1$ negative samples from the 'proposal' distribution $p(x_{t+k})$, we optimize:

$$\mathcal{L}_{\text{N}} = -\underset{X}{\mathbb{E}}\left[\log \frac{f_k(x_{t+k}, c_t)}{\sum_{x_j \in X} f_k(x_j, c_t)}\right] \tag{4}$$

Optimizing this loss will result in $f_k(x_{t+k}, c_t)$ estimating the density ratio in equation 2. This can be shown as follows.

The loss in Equation 4 is the categorical cross-entropy of classifying the positive sample correctly, with $\frac{f_k}{\sum_X f_k}$ being the prediction of the model. Let us write the optimal probability for this loss as $p(d = i|X, c_t)$ with $[d = i]$ being the indicator that sample x_i is the 'positive' sample. The probability that sample x_i was drawn from the conditional distribution $p(x_{t+k}|c_t)$ rather than the proposal distribution $p(x_{t+k})$ can be derived as follows:

$$\begin{aligned} p(d = i|X, c_t) &= \frac{p(x_i|c_t)\prod_{l\neq i} p(x_l)}{\sum_{j=1}^{N} p(x_j|c_t)\prod_{l\neq j} p(x_l)} \\ &= \frac{\frac{p(x_i|c_t)}{p(x_i)}}{\sum_{j=1}^{N} \frac{p(x_j|c_t)}{p(x_j)}}. \end{aligned} \tag{5}$$

As we can see, the optimal value for $f(x_{t+k}, c_t)$ in Equation 4 is proportional to $\frac{p(x_{t+k}|c_t)}{p(x_{t+k})}$ and this is independent of the the choice of the number of negative samples $N - 1$.

Though not required for training, we can evaluate the mutual information between the variables c_t and x_{t+k} as follows:

$$I(x_{t+k}, c_t) \geq \log(N) - \mathcal{L}_{\text{N}},$$

which becomes tighter as N becomes larger. Also observe that minimizing the InfoNCE loss $\mathcal{L}_{\text{N}}$ maximizes a lower bound on mutual information. For more details see Appendix.

2.4 Related Work

CPC is a new method that combines predicting future observations (predictive coding) with a probabilistic contrastive loss (Equation 4). This allows us to extract slow features, which maximize the mutual information of observations over long time horizons. Contrastive losses and predictive coding have individually been used in different ways before, which we will now discuss.

Contrastive loss functions have been used by many authors in the past. For example, the techniques proposed by [21, 22, 23] were based on triplet losses using a max-margin approach to separate positive from negative examples. More recent work includes Time Contrastive Networks [24] which proposes to minimize distances between embeddings from multiple viewpoints of the same scene and whilst maximizing distances between embeddings extracted from different timesteps. In Time Contrastive Learning [25] a contrastive loss is used to predict the segment-ID of multivariate time-series as a way to extract features and perform nonlinear ICA.

There has also been work and progress on defining prediction tasks from related observations as a way to extract useful representations, and many of these have been applied to language. In Word2Vec [9] neighbouring words are predicted using a contrastive loss. Skip-thought vectors [26] and Byte mLSTM [27] are alternatives which go beyond word prediction with a Recurrent Neural Network, and use maximum likelihood over sequences of observations. In Computer Vision [28] use a triplet loss on tracked video patches so that patches from the same object at different timesteps are more similar to each other than to random patches. [11, 29] propose to predict the relative postion of patches in an image and in [10] color values are predicted from a greyscale images.

3 Experiments

We present benchmarks on four different application domains: speech, images, natural language and reinforcement learning. For every domain we train CPC models and probe what the representations contain with either a linear classification task or qualitative evaluations, and in reinforcement learning we measure how the auxiliary CPC loss speeds up learning of the agent.

3.1 Audio

For audio, we use a 100-hour subset of the publicly available LibriSpeech dataset [30]. Although the dataset does not provide labels other than the raw text, we obtained force-aligned phone sequences